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United States Patent Application Publication

20250276458

Kind Code

A1

Publication Date

September 04, 2025

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KNIFE WITH PUSH BUTTON ARTICULATED SPRING

Abstract

A folding knife with a push button automatic open spring configuration to propel the pivoted knife blade outwardly from between the support handle frames. A bifurcated compound spring configuration has a transverse blade engagement portion and a pair of returning integral handle engagement arms against which the spring is compressed when in closed folded position within the handle. The bifurcated spring exerts opening force on the blade when the blade is released to a full open use position.

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Family ID: 96881634

Appl. No.: 18/592980

Filed: March 01, 2024

Publication Classification

Int. Cl.: B26B1/04 (20060101); B26B1/10 (20060101)

U.S. Cl.:

CPC B26B1/046 (20130101); B26B1/10 (20130101)

Background/Summary

BACKGROUND OF THE INVENTION

1. Technical Field

[0001] This invention relates to the field of folding knives, more specifically spring actuated blades

deploying the blade to an open useful full extended position automatically after release.

2. Description of Prior Art

[0002] Retractable knives that have folding pivoted blades that retract into the handle of the knife and have spring retainment and assist for deployment, see for example U.S. Pat. Nos. 6,105,255, 7,051,441, 7,676,932, 8,893,389, 9,707,687, 9,505,141 and 9,527,218.

[0003] U.S. Pat. No. 6,105,255 is directed to a folding knife having a leaf spring extending along the length of a registration slot in the compound handle.

[0004] U.S. Pat. No. 7,051,441 discloses a leaf spring assist open for a folding knife. The leaf spring defines an elongated arcuate configuration having a distal end to engage the tang of the blade and a shoulder engagement to tension the spring for blade deployment.

[0005] U.S. Pat. No. 8,893,389 is directed to a cantilever spring assist knife having a first end coupled to the handle and a second end adapted to act on the tang of the blade.

[0006] U.S. Pat. No. 7,676,932 illustrates a spring assisted blade in a folding knife. A spring assist closing force on a guide pin and the blade in closed position and on opening frames when blade is in partial open position.

[0007] U.S. Pat. No. 9,505,141 claims a folding knife with a manual mode and an automatic opening mode. A tension spring connected to pivot shaft enables opening in both modes.

[0008] U.S. Pat. No. 9,527,218 also shows dual operation modes with a spring assist.

[0009] U.S. Pat. No. 9,707,687 discloses an engagement mechanism which is switchable between engaged and disengaged condition in which the assisted opening mechanism is disabled so as not to provide opening assistance to the blade.

SUMMARY OF THE INVENTION

[0010] The present invention is dual bifurcated spring automatic opening actuation mechanism for a folding knife. A pivoted knife blade is retractable between a pair of handle frame and against the bifurcated spring engaging both handle frames and extending longitudinally and transversely against the edge at the base of the blade. A blade engagement lock holds the blade in closed position under spring tension until activation release of the blade and the engaged spring propels the hinge blade outwardly to open locked use position. The transverse engagement of the bifurcated spring on the blade assures stored energy transfer to the blade in a translateral manner.

Description

DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is an exploded perspective view of the folding knife and spring of the invention in open position.

[0012] FIG. 2 is an exploded perspective view of the folding knife and spring of the invention in closed position.

[0013] FIG. 3 is an enlarged top plan view of the folding knife in closed position.

[0014] FIG. 4 is a graphic sectional view of the folding knife with the spring of the invention in closed compressed position shown in solid lines and in open position in broken lines.

[0015] FIG. 5 is a perspective view of a folding knife in closed position prior to deployment of the blade.

[0016] FIG. 6 is an enlarged side elevational view of the knife's spring in rest position and in compressed position shown in broken lines within a spring retaining frame of the handle.

[0017] FIG. 7 is a top plan view thereof.

[0018] FIG. 8 is an end elevational view of the knife spring.

[0019] FIG. 9 is a side elevational sectional view of the folding knife and spring in open locked position.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Referring now to FIGS. 1 and 2 of the drawings, a dual spring **10** of the invention can be seen in a folding knife **11**. The folding knife **11** has a handle portion **12** adapted to receive a knife blade **13** pivotally positioned there within. The exploded view shows the handle portion **12** having a pair of oppositely disposed identical frame members **14** and **15**, each having multiple assembly apertures **A** aligned with one another for receiving tubular spacers **16** and corresponding fasteners **F** there through.

[0021] Spring receiving recesses **16A** and **16B** are formed respectively in each of the frame members **14** and **15** in aligned facing orientation to receive and hold the dual spring **10** of the invention there within. The folding knife blade **13** proximal end having an apertured contoured arcuate surface tang **13A** for receiving a blade pivot pin assembly **17** there through with aligned apertures in the corresponding frame members **14** and **15** have oppositely disposed spacer washers **W** and a pair of interengaging thread fasteners **F** for pivotally securing the blade **13** to the handle portion **12**, as will be well understood by those skilled in the art.

[0022] The proximal end of the knife blade **13** tang **13A** has a release engagement and retainment notches **18A** and **18B** aligned to receive a spring urged release button **19** as best seen in FIG. 5 of the drawings that is movable from a depressed release position to an engagement lock in both blade open and closed position.

[0023] As noted, the dual spring **10** of the invention is retained within the corresponding spring receiving and retaining recesses **16A** and **16B** that are of a general V-shape with a reduced transverse arcuate base end **17A** tapering outwardly to an oppositely disposed wider distal end **17B**.

[0024] Referring now to FIGS. 1-4 and 6-9 of the drawings, the dual spring **10** of the invention can be seen having a blade engagement portion **21** and a frame engagement portion **22**. The dual spring **10** is formed from a continuous length of contoured wire resilient material, in the example spring steel which under translateral compression will impart stored kinetic energy when released. The blade engagement portion **21** defines an elongated U-shaped configuration with a pair of spaced parallel extending arms **23** interconnected by an integral end linkage **24** that is in direct transverse contact with the knife blade **13** adjacent the blade retaining and release notches **18A** in the tang **13A** when folded in closed position with the spring **10** in compressed orientation as seen in FIGS. 2 and 5 of the drawings.

[0025] The spring frame engagement portion **22** has a pair of spaced parallel free end arms **25** that are interconnected to the respective corresponding blade engagement arms **22** in oppositely disposed relation thereto in a continuous configuration.

[0026] The frame engagement arms **25** are therefore registerable within the respective spring receiving recesses **16A** and **16B** as hereinbefore described when assembled.

[0027] It will be apparent that the spring force, force exerted by the spring perpendicular to the contact tangency line CTL on the knife blade, illustrated best in FIG. 6 of the drawings, supplies the opening force to the blade **13** once released sufficient to open the knife blade to a blade extended locked position as illustrated in FIG. 9 of the drawings.

[0028] In application, the dual spring **10** is assembled into the knife frame members **14** and **15** in aligned spring receiving recesses **16A** and **16B**, best seen in FIGS. 3, 4 and 6 of the drawings in a blade open configuration illustrated in FIGS. 1 and 9 of the drawings. Once assembled by corresponding interengaging fasteners **F**, the knife frames **14** and **15** secure the aforementioned dual spring **10** and folding knife blade **13** within the spring receiving area with a spring urged blade lock and release button **26** in positional orientation there through.

[0029] It will be seen in operation, the assembled release button **26** when depressed releases the knife blade **13** allowing it to move from an open locked use position to a folded lock position within the handle portion **12** thereby compressing the dual spring **10** within the spring receiving recesses **16A** and **16B** as seen in FIGS. 2, 5 and 6 of the drawings with the lock button **26** engaged in the corresponding notch locking the blade in place under spring tension.

[0030] For activation and automatic release, the lock button **26** is then depressed displacing the

button from a temporary locking engagement within the recess notch **18A** in the proximal end tang **13A** of the knife blade **13**.

[0031] The dual spring **10** in transverse linkage **24** engagement with the knife blade **13** surface **S** is released propelling the knife blade **13** from within the handle portion **12** to a fully open position as seen in FIGS. **1** and **9** of the drawings.

[0032] A blade stop pin **27** fixed to the handle frame engages the knife blade **13** upstanding stop portion **28** and the spring urged release button **26** registers in the locking notch **18B** of blade in the handle in open locked position.

[0033] Correspondingly, to close the knife **11**, the release button **26** is depressed displacing the button from the locking notch **18B** allowing the blade **13** to be folded into the handle portion **12** engaging and compressing the dual spring **10** as hereinbefore described.

[0034] It will be evident that the release button **26** automatically engages the tang notch **18A** locking the blade in closed folded position, as seen in FIG. **4** of the drawings.

[0035] It will be seen that by providing a bifurcated dual spring **10** in transverse end registration against the blade surface independently of the blade tang **13B** that a true auto opening “switch blade folding knife” can be achieved.

[0036] It will thus be seen that a new and novel bifurcated dual spring for application use within an automatic release folding knife has been illustrated and described and it will be apparent to those skilled in the art that various changes and modifications may be made thereto without departing from the spirit of the invention.

Claims

1. An automatic opening folding blade knife comprising, a handle portion defining a recess into which a knife blade is capable of retracting, A bifurcated dual blade engagement spring retained within said handle portion, Said bifurcated dual blade engagement spring having elongated pairs of spaced parallel arms, said arm pairs interconnected to one another defining a frame engagement arm pair and a blade engagement arm pair, blade engagement arm pair having interconnected end linkage for direct transverse contact with said knife blade when retracted within the handle recess, a blade release, registerable with said knife blade.
 2. The automatic opening folding knife blade of claim 1 wherein said knife blade has an apertured proximal end tang with locking and release notches therein.
 3. The automatic opening folding knife blade of claim 1 wherein said bifurcated dual blade engagement spring applies force on said blade adjacent to said blade release.
 4. The automatic opening folding knife blade of claim 1 wherein said bifurcated dual blade engagement spring frame engagement arm pairs have respective free ends in contact with said handle portion.
 5. The automatic opening folding knife blade of claim 1 wherein said bifurcated dual blade engagement spring pairs of spaced parallel arms further comprises, a single continuous spring wire.
 6. The automatic opening folding knife blade of claim 1 wherein said blade release registerable with said knife blade comprises, a spring urged button extending through said frame and said handle portion.
 7. The automatic opening folding knife blade of claim 1 wherein said spring pairs of spaced parallel arms are in spaced vertical overlying position to one another.
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